```
> restart;# The modified Planck's law
> restart;#Planck  for γ - ray
> λ2 := 345.35·10^-9; λ1 := 7.8·10^-10; c := 299792458; v1 := c/λ1; E1 := 3.59; E3 := 2.446
      ·10^6 ; λ3 := E1/E3·7.8·10^-10; v3 := c/λ3; rb := 5.292410932 10^-11;
```

$$\lambda 2 := 3.453500000 \times 10^{-7}$$

$$\lambda 1 := 7.800000000 \times 10^{-10}$$

$$c := 299792458$$

$$v1 := 3.843493051 \times 10^{17}$$

$$E1 := 3.59$$

$$E3 := 2.446000000 \times 10^{6}$$

$$\lambda 3 := 1.144807849 \times 10^{-15}$$

$$v3 := 2.618714208 \times 10^{23}$$

$$rb := 5.292410932 \times 10^{-11} \quad (1)$$

```
> Eγ := diff( Tq·32·π^2·ν^2/(2·3)·E0, nu ); ; Tq := 1.765358264×10^-19#The  calculus  = 8·Pi^2
      ·ν^2/c^2  4·π·rb^3/(3·Tq) ·U(ν, t) thus for γ −ray  the internal energy  U(ν,
      t) is constant #'E0
      ' because the energy of  a transversal wave  is constant like  the
      initial oscillation of  one  nucleus
      and depend only of  number of states 32·Pi^2·rb^3·ν^2/(3·c^2·Tq);
```

$$E\gamma := \frac{32\, Tq\, \pi^2\, \nu\, E0}{3}$$

$$Tq := 1.765358264 \times 10^{-19} \quad (2)$$

```
>
> Eγ := 32  Tq  π^2  ν  E0/3;
```

$$E\gamma := 1.858494688 \times 10^{-17}\, \nu\, E0 \quad (3)$$

```
> c := 299792458;  Sq := 4.25·10^-19;   h := 4.135667697×10^-15; R := 3.566639512
      ×10^22; Tq := 1.765358264×10^-19; ec := 1.602176634×10^-19; v3 := c/λ3;  v1 := c/λ1;
      #Vg := R/(2  Tq) ;
```

$$c := 299792458$$

$$Sq := 4.250000000 \times 10^{-19}$$

$$h := 4.135667697 \times 10^{-15}$$

$$R := 3.566639512 \times 10^{22}$$

$$Tq := 1.765358264 \times 10^{-19}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$v3 := 2.618714208 \times 10^{23}$$

$$v1 := 3.843493051 \times 10^{17} \quad (4)$$

```
> Eγ := 32 Tq π^2 ν E0 / 3;
```

$$E\gamma := 1.858494688 \times 10^{-17}\, \nu\, E0 \quad (5)$$

```
> λmin := evalf(c / 10^23);
    # Thus the minimum wavelenght is ~ times the nucleon radius as in plot
    below
```

$$\lambda min := 2.997924580 \times 10^{-15} \quad (6)$$

```
> hγ := E1 / v1;
```

$$h\gamma := 9.340461794 \times 10^{-18} \quad (7)$$

```
> solve(3.59 = g·v1); hγ := 9.340461795×10^-18; subs(ν = 3.843493051×10^17, hγ·ν);
    solve(hγ = 32 Tq π^2 E0 / 3, E0)
```

$$9.340461794 \times 10^{-18}$$

$$h\gamma := 9.340461795 \times 10^{-18}$$

$$3.590000000$$

$$0.5025821088 \quad (8)$$

```
> E1 := 1239.84 / (λ2·10^9);
    # The Energy of  3.59eV for photons of  wave-lenght 345.35 nm;
```

$$E1 := 3.590097003 \quad (9)$$

```
> E3 := 2.446·10^6 ; λ3 := E1/E3·7.8·10^-10; v3 := c/λ3; λ1 := 7.8·10^-10; v1 := c/λ1;
```

$$E3 := 2.446000000 \times 10^{6}$$

$$\lambda 3 := 1.144838783 \times 10^{-15}$$

$$v3 := 2.618643450 \times 10^{23}$$

$$\lambda 1 := 7.800000000 \times 10^{-10}$$

$$v1 := 3.843493051 \times 10^{17} \quad (10)$$

```
> E:=he*nu;
```

$$E := he\,\nu \quad (11)$$

```
>  #The Planck 's  law for γ−ray :  hγ  =9.340461795×10^-18  in eV·s ;
> hγ  := 9.3404617950×10^-18; ec := 1.602176634·10^-19;  c := 299792458; h :=
```

$$\frac{6.62607015\cdot10^{-34}}{ec};\ he := 4.13566733\cdot10^{-15};\ \lambda1 := 7.8\cdot10^{-10}\ ;\ v1 := \frac{c}{7.8\cdot10^{-10}}$$

```
# h  4.135667697×10^-15eV·s & hγ = 9.3404617950×10^-18eV·s;
```

$$h\gamma := 9.340461795 \times 10^{-18}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$c := 299792458$$

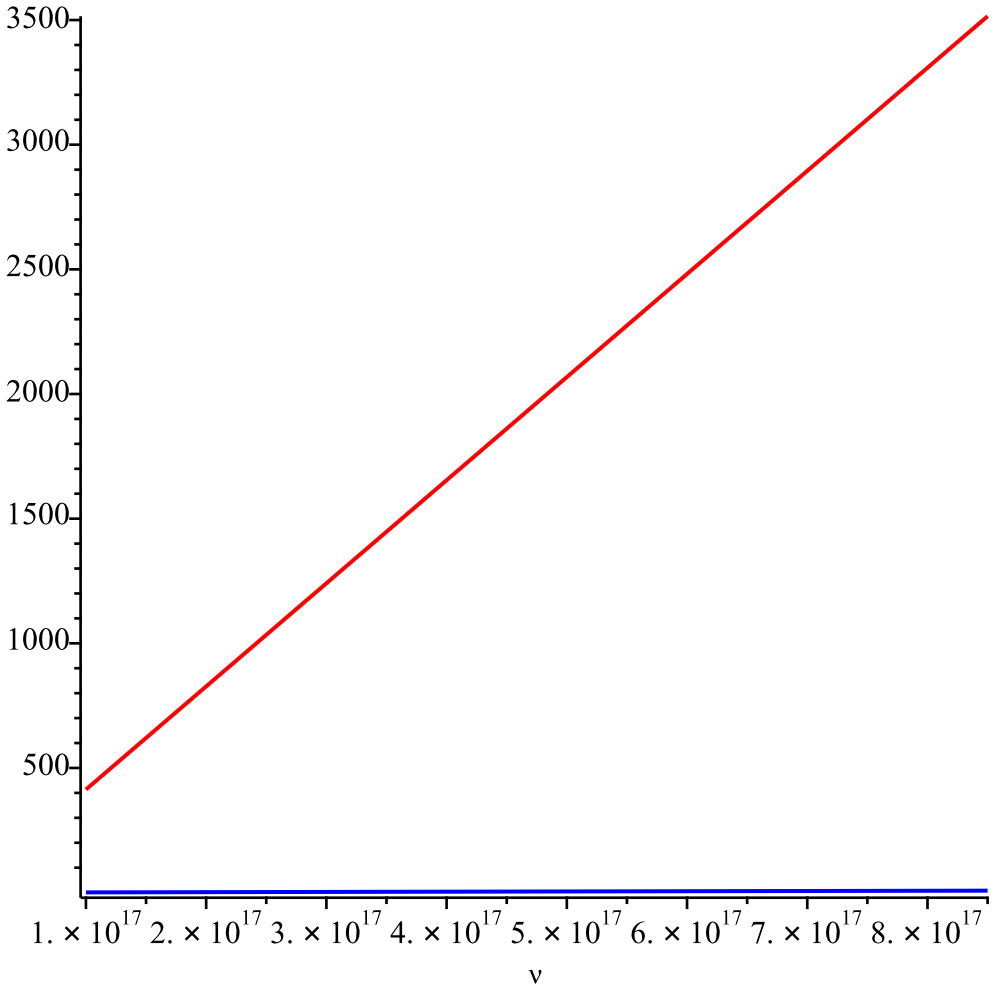
$$h := 4.135667697 \times 10^{-15}$$

$$he := 4.135667330 \times 10^{-15}$$

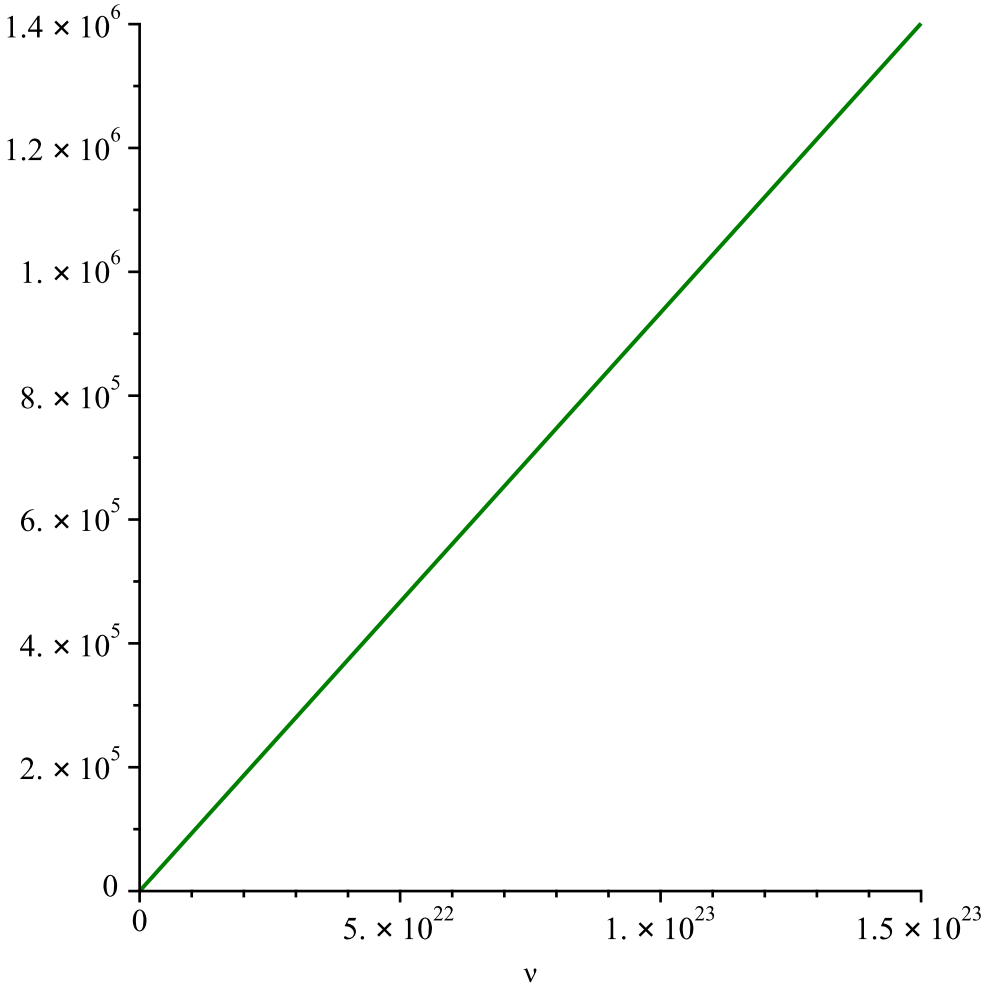
$$\lambda1 := 7.800000000 \times 10^{-10}$$

$$v1 := 3.843493051 \times 10^{17} \quad \textbf{(12)}$$

```
> plot([9.3404617950×10^-18·ν, E], ν= 10^17 .. 8.5·10^17, color=["Blue", "Red"]);
  #The Planck's law for γ - ray liniar approx. "Blue" the real energy
  and Red with photon Planck constant which does
  not match the real energies!!
```




```
> plot(9.3404617950×10^-18·ν, ν= 3·10^17 ... 15·10^22, color = "Green" );
  #The Planck's law for γ - ray "Green" with` `γ − ray Planck liniar the
  real value of energy
```




```
> #  Thus, is a huge difference in the calculus of the gamma-ray energy with
  the two equations.
> restart;#The best fit for photons Planck's low
> E(nu):=Em+E0*ln(1+2.787176283*10^(-17)*nu)+C*nu^2;
```

$$E(\nu) := Em + E0 \ln\left(1 + 2.787176283 \times 10^{-17} \nu\right) + C \nu^2 \quad (13)$$

```
> E1:=subs(nu=9.733521364*10^16,E(nu));
```

$$E1 := Em + E0 \ln\left(3.712903990\right) + 9.474143814 \times 10^{33} C \quad (14)$$

```
> E2:=subs(nu=7.880979442*10^14,E(nu));
```

$$E2 := Em + E0 \ln\left(1.021965679\right) + 6.210983697 \times 10^{29} C \quad (15)$$

```
> E3:=subs(nu=8.689636465*10^14,E(nu));
```

$$E3 := Em + E0 \ln\left(1.024219549\right) + 7.550978189 \times 10^{29} C \quad (16)$$

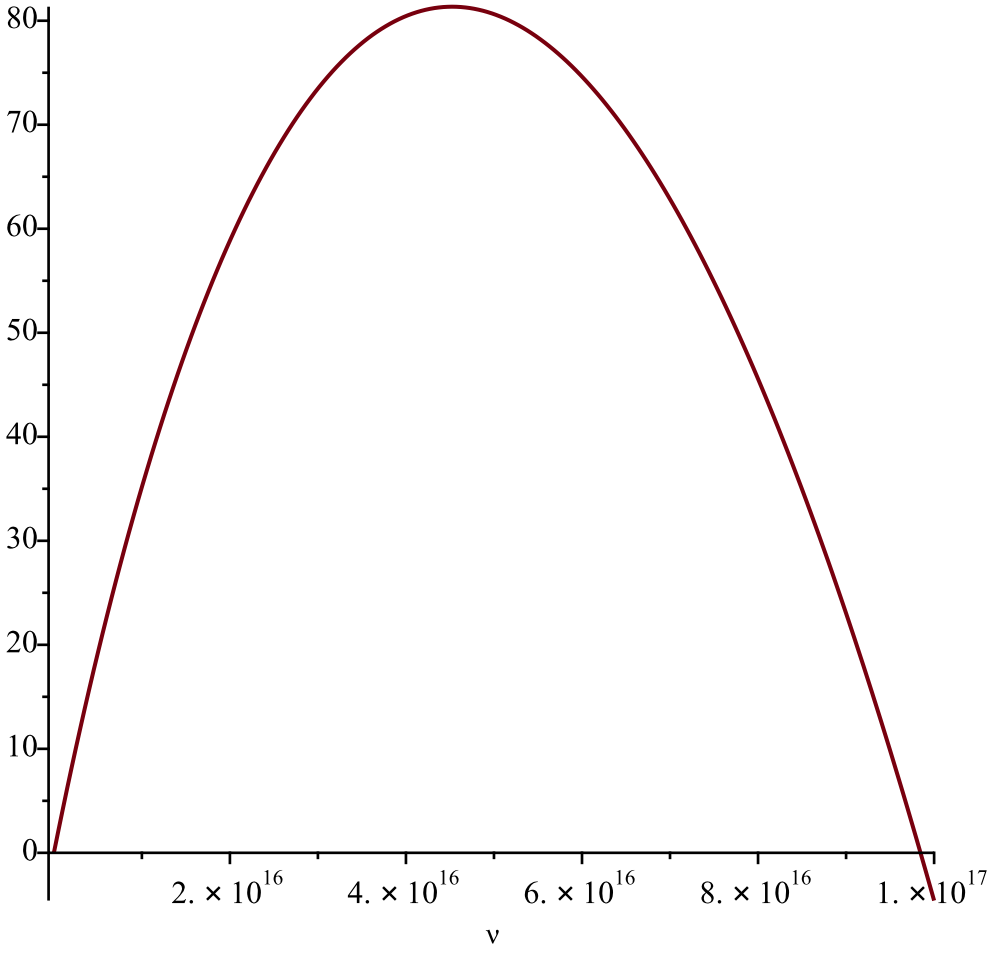
```
> solve({E1=3.259,E2=3.259,E3=3.59});
```

$$\left\{C = -2.063162888 \times 10^{-32}, E0 = 151.5047392, Em = -0.02006692051\right\} \quad (17)$$

```
> E(nu):=subs(Em=-0.02006692051,E0=151.5047392,C=-2.063162888*10^(-32),E(nu)
  );
```

$$E(\nu) := -0.02006692051 + 151.5047392 \ln\left(1 + 2.787176283 \times 10^{-17} \nu\right) - 2.063162888 \times 10^{-32} \nu^2 \quad (18)$$

```
> plot(E(nu),nu=10^13..10^17, caption = typeset("Graph of the Planck's
  constant [eV] as a function of λ in [nm] "));
```



Graph of the Planck's constant [eV] as a function of λ in [nm]

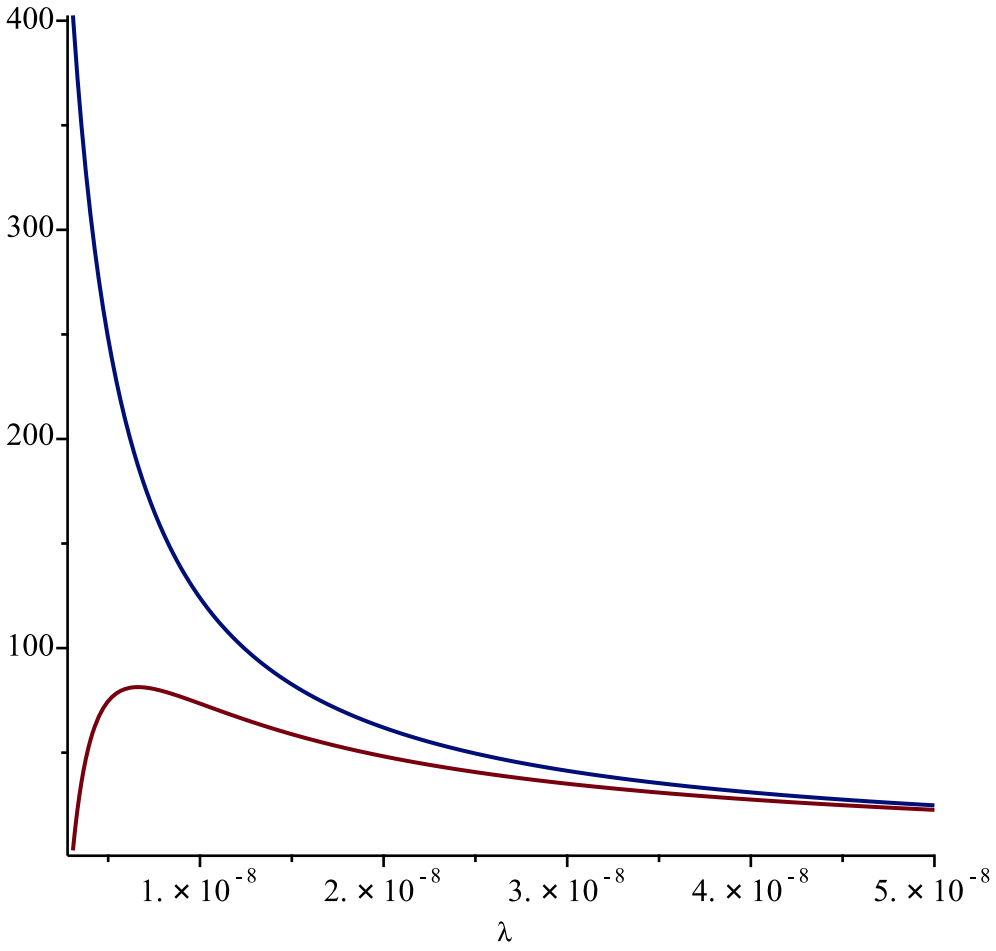
```
> he:=6.6260701500*10^(-34)/1.6021766340/10^(-19);
```

$$he := 4.135667697 \times 10^{-15} \tag{19}$$

```
> c := 299792458;B := lambda -> subs(nu = c/lambda, -0.02006692051 +
  151.5047392*ln(1 + 2.787176283*1/100000000000000000*nu) + (-1)*
  2.063162888*1/10000000000000000000000000000000*nu^2);plot([B(lambda),he*
  c/lambda], lambda = 3.08*10^(-9) .. 5*10^(-8),caption = typeset("A plot of
  Energy theoretical and practical in [eV] versus lambda in [nm]"));
```

$$c := 299792458$$

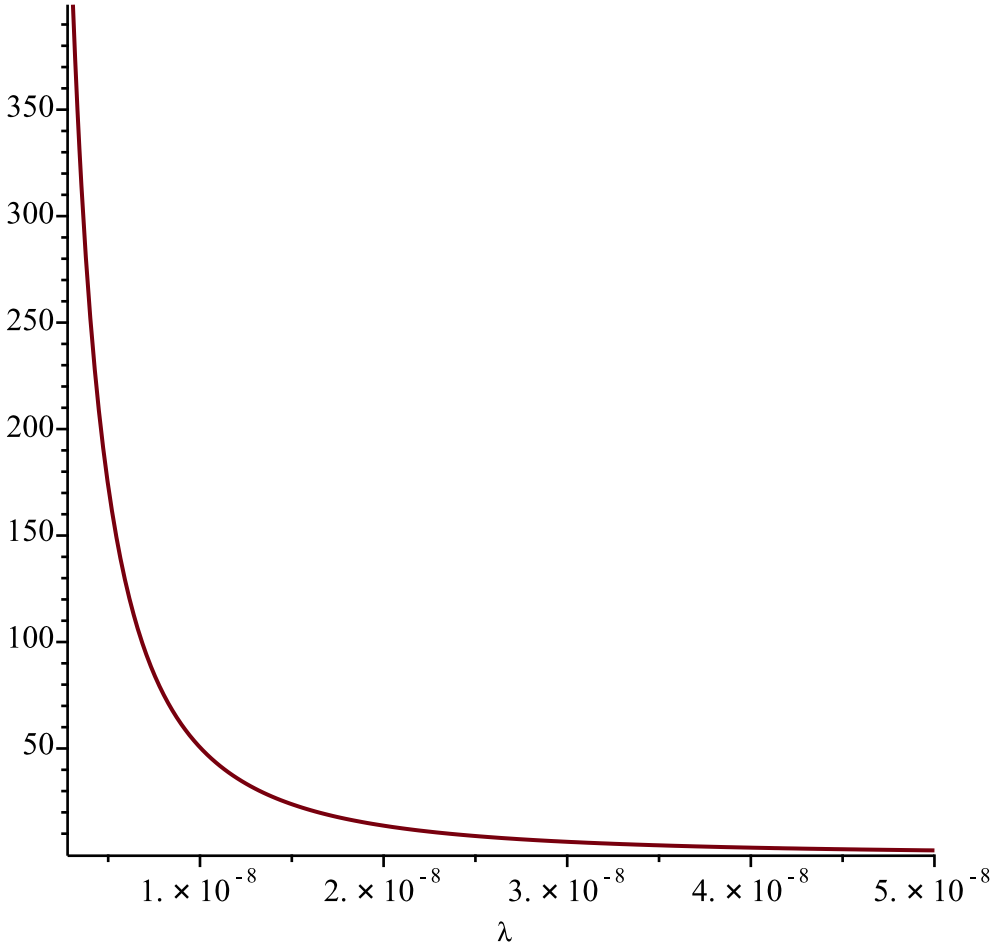
$$B := \lambda \mapsto subs\left(\nu = \frac{c}{\lambda}, -0.02006692051 + 151.5047392 \cdot \ln\left(1 + \frac{2.787176283 \cdot 1 \cdot \nu}{100000000000000000}\right) + \frac{(-1) \cdot 2.063162888 \cdot 1 \cdot \nu^2}{10000000000000000000000000000000000}\right)$$



A plot of Energy theoretical and practical in [eV] versus lambda in [nm]

```
> -B(lambda)+he*c/lambda;plot(%,lambda = 3.08*10^(-9) .. 5*10^(-8), caption
  = typeset("Graph of the error [eV] as a function of λ in [nm] between the
  two calculation methods "));# Below the wavelength of 30 nm, the error
  becomes larger and below 15-20nm the error is dominant.
```

$$0.02006692051 - 151.5047392 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) + \frac{1.854278330 \times 10^{-15}}{\lambda^2} + \frac{1.239841984 \times 10^{-6}}{\lambda}$$



Graph of the error [eV] as a function of λ in [nm] between the two calculation

```
> solve(diff(B(lambda), lambda) = 0);# @ lambda= 6.625nm the photons have
  maximum of energy
```

$$-3.695060778 \times 10^{-9},\ 6.624561165 \times 10^{-9} \tag{20}$$

```
> h := 6.6260701500*10^(-34);ec := 1.6021766340*10^(-19);rb:=5.239*10^(-13);
  he:=h/ec;
  c := 299792458;T := 273 + 24;k := 1.3806490000*10^(-23);Tq:= 1.765*10^
  (-19);nu0:=evalf(c/(3.804*10^(-7)));nu1:=evalf(c/(3.08*10^(-9)));nu2:=
  evalf(c/(3.135*10^(-7)))
```

$$h := 6.626070150 \times 10^{-34}$$

$$ec := 1.602176634 \times 10^{-19}$$

$$rb := 5.239000000 \times 10^{-13}$$

$$he := 4.135667697 \times 10^{-15}$$

$$c := 299792458$$

$$T := 297$$

$$k := 1.380649000 \times 10^{-23}$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$v0 := 7.880979442 \times 10^{14}$$

$$v1 := 9.733521364 \times 10^{16}$$

$$v2 := 9.562757830 \times 10^{14} \tag{21}$$

```
> evalf(subs(nu=7.880979442*10^14,E(nu)));evalf(subs(nu=c/(380.43*10^(-9)),E
```

```
(nu)));
```

3.258999999

3.258745253 (22)

```
> evalf(subs(nu=8.689636465*10^14,E(nu)));evalf(subs(nu=c/(345.35*10^(-9)),E
(nu)));
```

3.590000000

3.586400619 (23)

```
> evalf(subs(nu=9.733521364*10^16,E(nu)));nu1:=evalf(c/(3.08*10^(-9)));
```

3.2590000

$$v1 := 9.733521364 \times 10^{16} \quad (24)$$

```
> evalf(subs(nu=nu2,E(nu)));evalf(subs(nu=c/(17*10^(-9)),E(nu)));
```

3.946259923

54.13412164 (25)

```
> solve(-0.02006692051 + 151.5047392*ln(1 + 2.787176283*10^(-17)*nu) -
2.063162888*10^(-32)*nu^2=5.28,nu)
```

Warning, solutions may have been lost

```
> evalf(subs(nu=5*10^16,E(nu)));# ~The Energy peak in eV
```

80.63322078 (26)

```
> evalf(subs(nu=0.984563*10^17,E(nu)));c/(0.984563*10^17);
```

0.0000560

$$3.044929150 \times 10^{-9} \quad (27)$$

```
> 1239.84/8.54;nu3:=c/(%*10^(-9));
```

145.1803279

$$v3 := 2.064966117 \times 10^{15} \quad (28)$$

```
> evalf(subs(nu=nu3,E(nu)));8.54-%;# @ 145nm the error is 0.16eV;
```

8.370000712

0.169999288 (29)

```
> 1239.84/5.32;nu3:=c/(%*10^(-9));
```

233.0526316

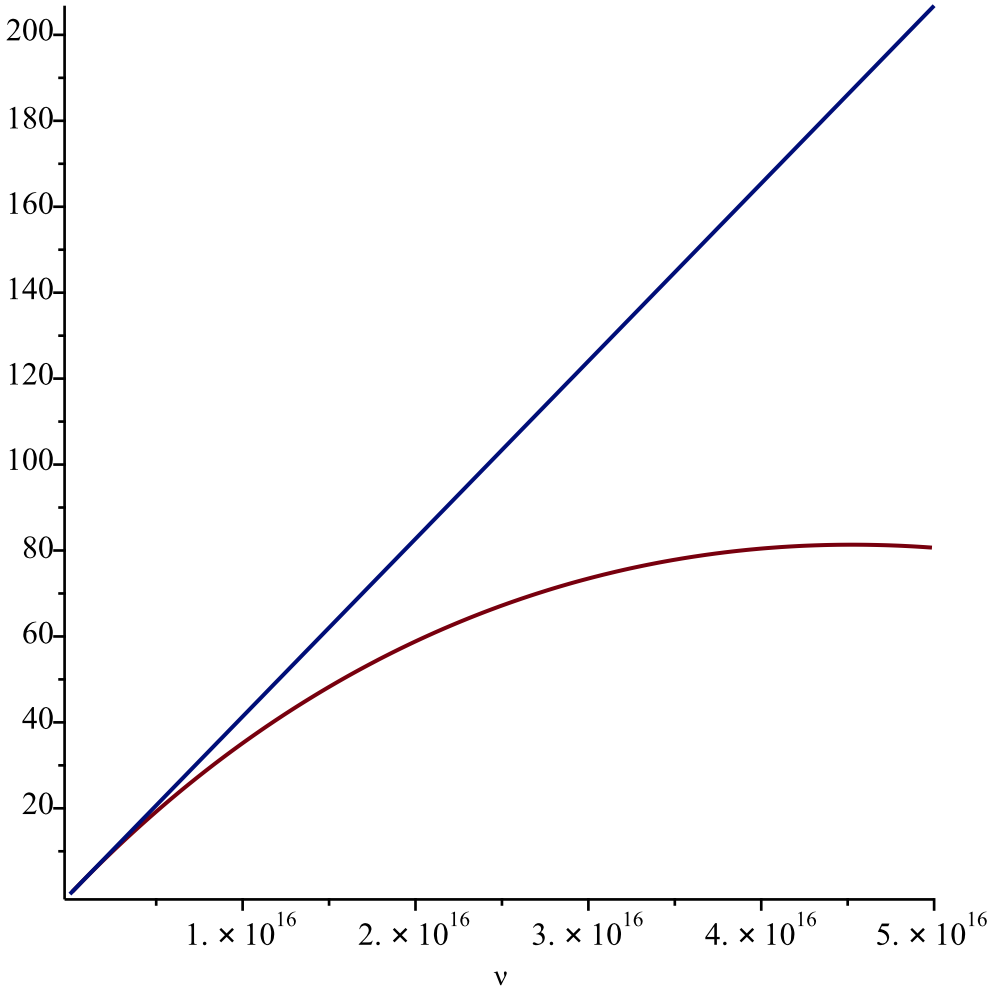
$$v3 := 1.286372335 \times 10^{15} \quad (30)$$

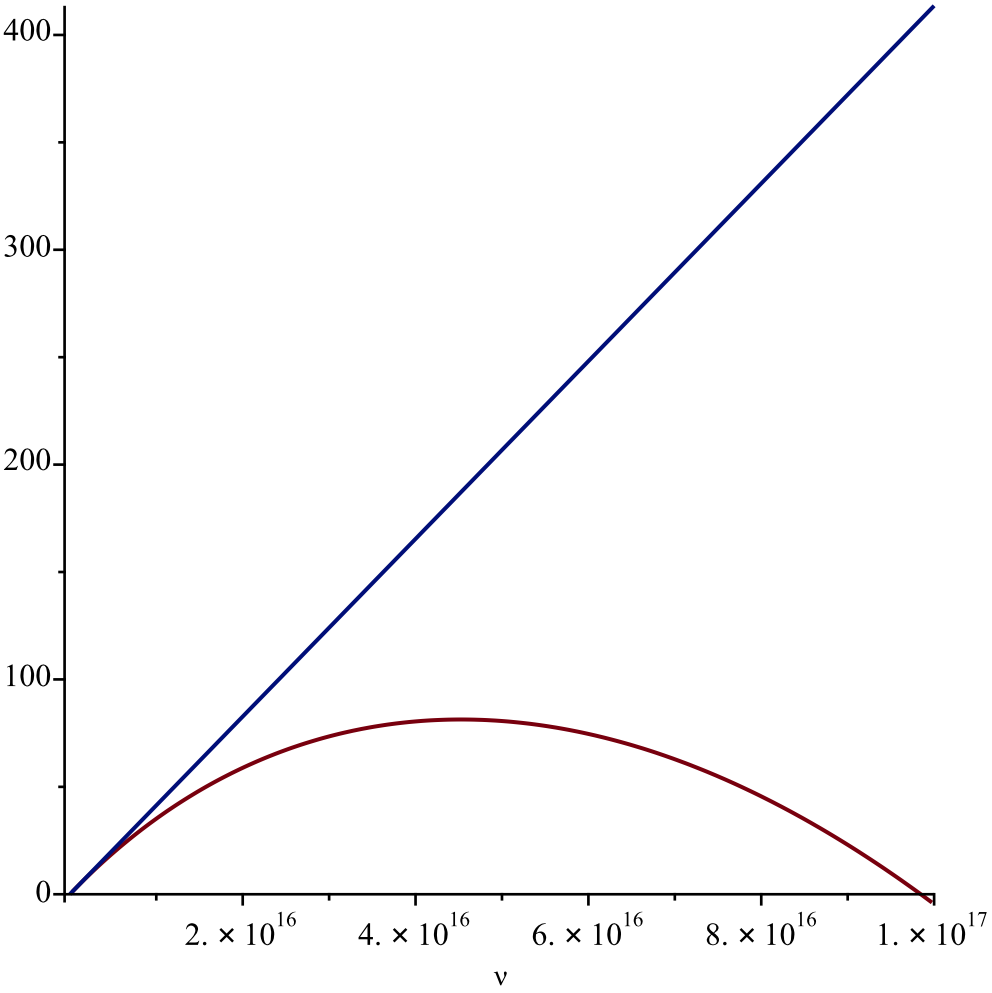
```
> evalf(subs(nu=nu3,E(nu)));5.32-%;#@ 233nm the error is 0.037eV;
```

5.282651909

0.037348091 (31)

```
> plot([E(nu),he*nu], nu=10^13..5*10^16);plot([E(nu),he*nu], nu=10^13.
.10^17);
```



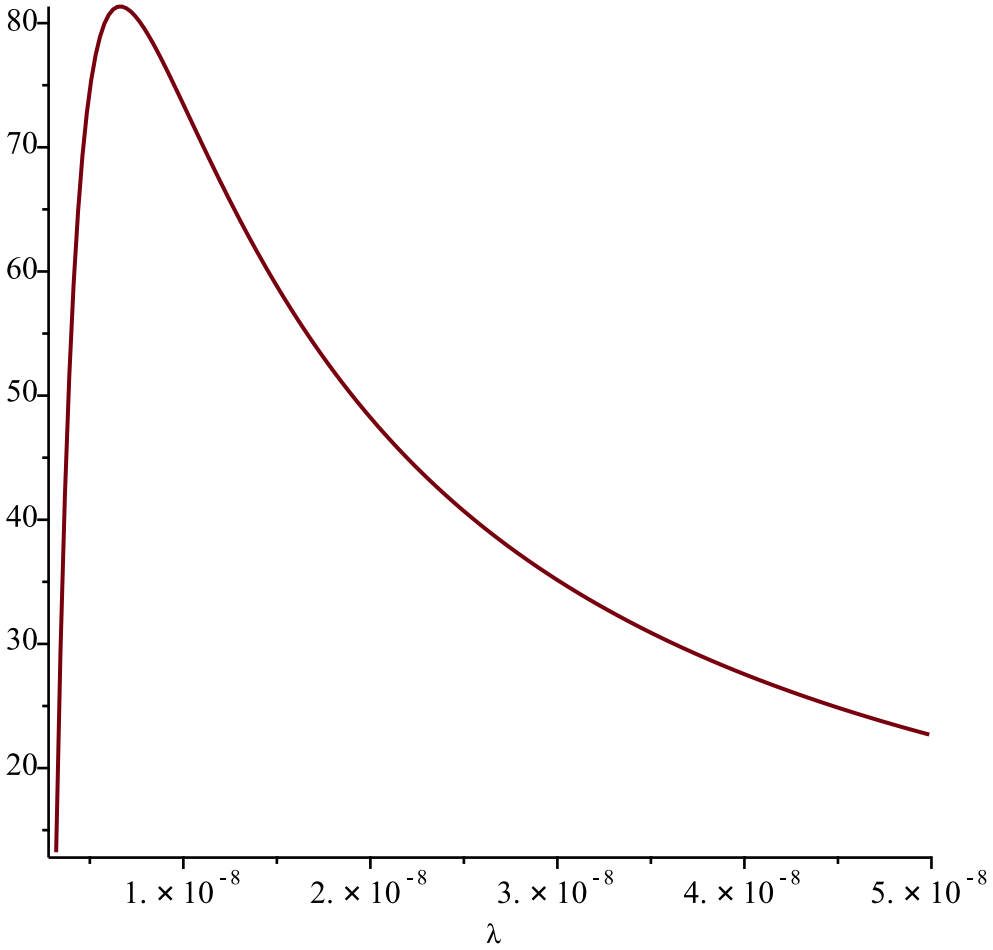



```
> c := 299792458;B := lambda -> subs(nu = c/lambda, -0.02006692051 +
```

```
151.5047392*ln(1 + 2.787176283*1/10000000000000000*nu) + (-1)*
2.063162888*1/10000000000000000000000000000000*nu^2);plot(B(lambda),
lambda = 3.08*10^(-9) .. 5*10^(-8),caption = typeset("A plot of Energy
[eV] versus lambda in [nm]"));
```

$$c := 299792458$$

$$B := \lambda \mapsto subs\left(\nu = \frac{c}{\lambda},\ -0.02006692051 + 151.5047392 \cdot \ln\left(1 + 2.787176283 \times 10^{-17} \cdot \nu\right) - 2.063162888 \times 10^{-32} \cdot \nu^{2}\right)$$



A plot of Energy [eV] versus lambda in [nm]

```
> B(lambda);
```

$$-0.02006692051 + 151.5047392 \ln\left(1 + \frac{8.355744288 \times 10^{-9}}{\lambda}\right) - \frac{1.854278330 \times 10^{-15}}{\lambda^{2}} \quad (32)$$

```
> solve(diff(B(lambda), lambda) = 0);# @ lambda= 6.625nm the photons have
maximum of energy
```

$$-3.695060778 \times 10^{-9},\ 6.624561165 \times 10^{-9} \quad (33)$$

```
>
> restart;#The Phonon Planck's "constant"
> Digits:=20;
```

$$Digits := 20 \quad (34)$$

```
> E(nu):=Em+E0*ln(1+2*Tq1*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln\left(2\ Tq1\ A\ \nu + 1\right) + C\ \nu^{2} \quad (35)$$

```
> c:=299792458;nu1:=435;ec:= 1.602176634*10^(-19);E1:=3.259;T:=297;va:=343;
  Tq1:=c/va*1.765*10^(-19);ha:=0.007486790719; ha*ec;Tq1*105.2757803;2*Tq1*
  32/3*Pi^2;
```

$$c := 299792458$$

$$v1 := 435$$

$$ec := 1.6021766340000000000 \times 10^{-19}$$

$$E1 := 3.259$$

$$T := 297$$

$$va := 343$$

$$Tq1 := 1.5426638144897959184 \times 10^{-13}$$

$$ha := 0.007486790719$$

$$1.1995161153629859846 \times 10^{-21}$$

$$1.6240513681098771170 \times 10^{-11}$$

$$3.2481027355498213704 \times 10^{-11} \quad (36)$$

```
> E(nu):=Em+E0*ln(1+A*3.248102735549821 3704*10^(-11)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln\left(1 + 3.2481027355498213704 \times 10^{-11} A \nu\right) + C \nu^2 \quad (37)$$

```
> E1:=subs(nu=435,E(nu));
```

$$E1 := Em + E0 \ln\left(1 + 1.4129246899641722961 \times 10^{-8} A\right) + 189225\, C \quad (38)$$

```
> E2:=subs(nu=528,E(nu));
```

$$E2 := Em + E0 \ln\left(1 + 1.7149982443703056836 \times 10^{-8} A\right) + 278784\, C \quad (39)$$

```
> E3:=subs(nu=25000,E(nu));
```

$$E3 := Em + E0 \ln\left(1 + 8.1202568388745534260 \times 10^{-7} A\right) + 625000000\, C \quad (40)$$

```
> solve({E1=3.259,E2=3.95});
```

$$\{A = A,\ C = 0.00001116582364698131957 7\, E0 \ln\left(1. + 1.4129246899641722961 \times 10^{-8} A\right) \quad (41)$$
$$- 0.00001116582364698131957 7\, E0 \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right)$$
$$+ 7.7155841400640918277 \times 10^{-6},\ E0 = E0,\ Em = 2.1128529796000401970\, E0 \ln\left(1.\right.$$
$$\left. + 1.7149982443703056836 \times 10^{-8} A\right) - 3.1128529796000401970\, E0 \ln\left(1. + 1.4129246899641722961\right.$$
$$\left. \times 10^{-8} A\right) + 1.7990185910963722239\}$$

```
> sol:=solve({E1=3.259,E2=3.95,E3=0},{Em,E0,C,A});
```

$$sol := \left\{A = A,\ C = -\left(0.0010000000000000000000\left(-3259. \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right)\right.\right.\right. \quad (42)$$
$$- 691. \ln\left(1. + 8.1202568388745534260 \times 10^{-7} A\right) + 3950. \ln\left(1. + 1.4129246899641722961\right.$$
$$\left.\left.\left. \times 10^{-8} A\right)\right)\right) \Big/ \left(89559. \ln\left(1. + 8.1202568388745534260 \times 10^{-7} A\right) - 6.24810775 \times 10^{8} \ln\left(1.\right.\right.$$
$$\left.\left. + 1.7149982443703056836 \times 10^{-8} A\right) + 6.24721216 \times 10^{8} \ln\left(1. + 1.4129246899641722961\right.\right.$$

$$\times 10^{-8} A)\Big), E0 = -\left(4.3203611830600000000 \times 10^{8}\right) \Big/ \left(89559.\ln\left(1. + 8.1202568388745534260\right.\right.$$

$$\times 10^{-7} A) - 6.24810775 \times 10^{8} \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right) + 6.24721216 \times 10^{8} \ln\left(1.\right.$$

$$\left.\left. + 1.4129246899641722961 \times 10^{-8} A\right)\right), Em = \left(0.002000000000000000000 \left(8.0559153 \times 10^{7} \ln\left(1.\right.\right.\right.$$

$$+ 8.1202568388745534260 \times 10^{-7} A) - 1.018437500000 \times 10^{12} \ln\left(1. + 1.7149982443703056836\right.$$

$$\left.\left.\left.\times 10^{-8} A\right) + 1.234375000000 \times 10^{12} \ln\left(1. + 1.4129246899641722961 \times 10^{-8} A\right)\right)\right) \Big/ \left(89559.\ln\left(1.\right.\right.$$

$$\left.\left. + 8.1202568388745534260 \times 10^{-7} A) - 6.24810775 \times 10^{8} \ln\left(1. + 1.7149982443703056836 \times 10^{-8} A\right)\right.\right.$$

$$\left.\left. + 6.24721216 \times 10^{8} \ln\left(1. + 1.4129246899641722961 \times 10^{-8} A\right)\right)\right\}$$

```
> A:=9.949747*10^6;evalf(sol[4]);#Em
```

$$A := 9.949747000000 \times 10^{6}$$

$$Em = -0.26049922890091899376 \quad (43)$$

```
> Em := -0.26049922890091899376;Em;
```

$$Em := -0.26219418286029721944$$

$$-0.26219418286029721944 \quad (44)$$

```
> evalf(sol[3]);#E0
```

$$E0 = 26.892256305433408378 \quad (45)$$

```
> E0 := 26.892256305433408378;E0;
```

$$E0 := 26.914388251007721676$$

$$26.914388251007721676 \quad (46)$$

```
> evalf(sol[2]);;#C
```

$$C = -9.4502695574974682733 \times 10^{-8} \quad (47)$$

```
> C:=-9.4502695574974682733*10^(-8);
```

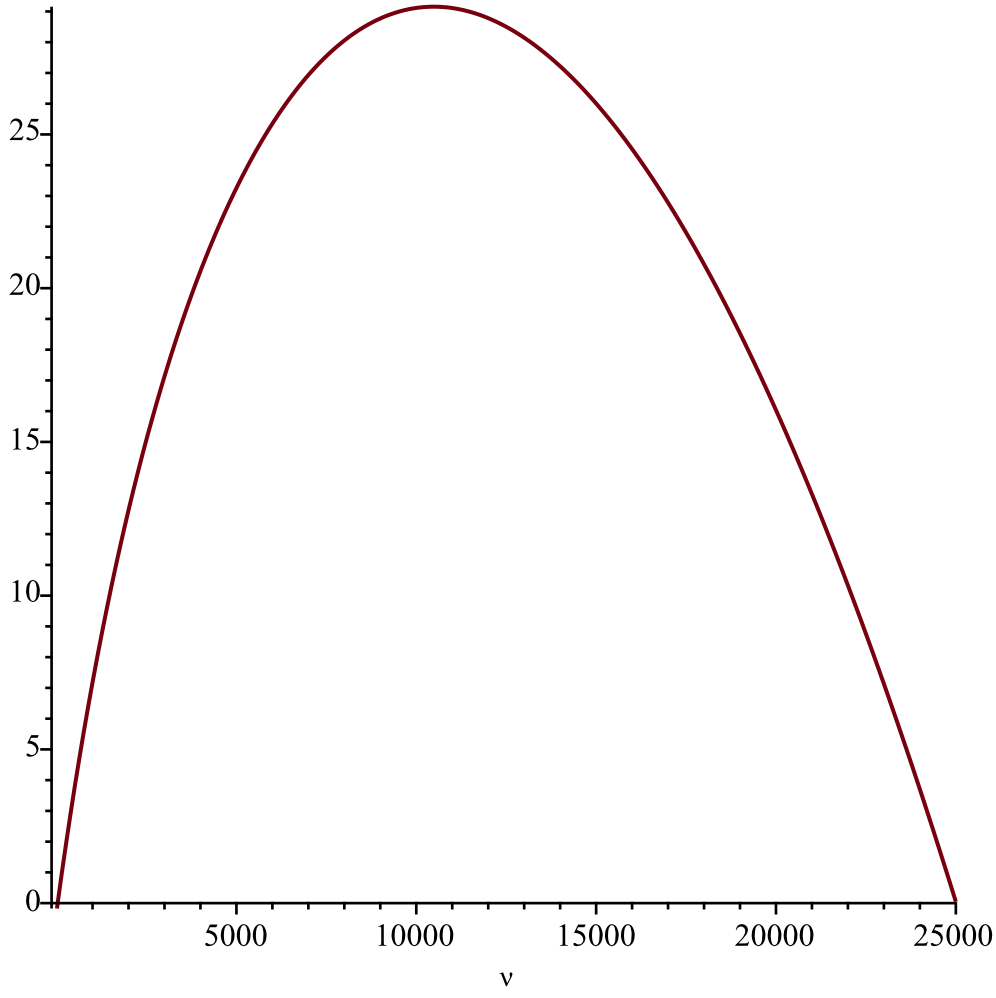
$$C := -9.4502695574974682733 \times 10^{-8} \quad (48)$$

```
> E(nu):=Em+E0*ln(1+A*3.2481027355498213704*10^(-11)*nu)+ C*nu^2;
```

$$E(\nu) := -0.26219418286029721944 + 26.914388251007721676 \ln\left(1 + 0.00032317800448728628531\ \nu\right) \quad (49)$$

$$- 9.4502695574974682733 \times 10^{-8} \nu^{2}$$

```
> plot(E(nu),nu=10..25000);
```




```
> evalf(subs(nu=435,E(nu)));evalf(subs(nu=528,E(nu)));evalf(subs(nu=480,E
  (nu)));
```

$$3.2602162608519398957$$

$$3.9517919094482611277$$

$$3.5973279580401586754 \quad (50)$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=30,E(nu)));evalf(subs(nu=13350,E(nu)
  ));#The max value is ~26.7eV at frequency ~13350Hz [30Hz÷24khz]
```

$$-0.26219418286029721944$$

$$-0.00259194059505890622$$

$$27.853768233661099002 \quad (51)$$

```
> restart;#The momopole RadioFrequency Planck's "constant"
> Digits:=25;
```

$$Digits := 25 \quad (52)$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, A\, Tq\, \nu + 1) + C\, \nu^2 \quad (53)$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=3.693*10^(-18);ec:=
  1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 3.693000000000000000000000 \times 10^{-18}$$

$$ec := 1.60217663400000000000000 \times 10^{-19} \quad (54)$$

```
> A0:=64/3*Tq;
```

$$A0 := 3.76533333333333333333333 \times 10^{-18} \quad (55)$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln\left(1 + 3.76533333333333333300000 \times 10^{-18} A \nu\right) + C \nu^2 \quad (56)$$

```
> E1:=subs(nu=5.754173859*10^8,E(nu));
```

$$E1 := Em + E0 \ln\left(1 + 2.16663826370879999998081 9 \times 10^{-9} A\right) + 3.31105167995989518810000 0 \times 10^{17} C \quad (57)$$

```
> E2:=subs(nu=6.39691958*10^8,E(nu));
```

$$E2 := Em + E0 \ln\left(1 + 2.40865345252266666664534 4 \times 10^{-9} A\right) + 4.09205801129873764000000 0 \times 10^{17} C \quad (58)$$

```
> E3:=subs(nu=31*10^9,E(nu));
```

$$E3 := Em + E0 \ln\left(1 + 1.16725333333333333323000 \times 10^{-7} A\right) + 961000000000000000000 C \quad (59)$$

```
> solve({E1=3.259,E2=3.59});
```

$$\{A = A, C = 1.28039935129046520417716 0 \times 10^{-17} E0 \ln\left(1. + 2.16663826370879999998081 9 \times 10^{-9} A\right) \quad (60)$$
$$- 1.28039935129046520417716 0 \times 10^{-17} E0 \ln\left(1. + 2.40865345252266666664534 4 \times 10^{-9} A\right)$$
$$+ 4.23812185277143982582639 8 \times 10^{-18}, E0 = E0, Em = 4.23946842310985480713194 1 E0 \ln\left(1.$$
$$+ 2.40865345252266666664534 4 \times 10^{-9} A\right) - 5.23946842310985480713194 1 E0 \ln\left(1.$$
$$+ 2.16663826370879999998081 9 \times 10^{-9} A\right) + 1.85573595195063805883932 7\}$$

```
> sol:=solve({E1=3.259,E2=3.59,E3=0},{Em,E0,C,A});
```

$$sol := \left\{A = A, C = -\left(0.10000000000000000000000 0\left(-3259. \ln\left(1. + 2.40865345252266666664534 4\right.\right.\right. \quad (61)$$
$$\left.\times 10^{-9} A\right) - 331. \ln\left(1. + 1.16725333333333333323000 \times 10^{-7} A\right) + 3590. \ln\left(1.$$
$$\left.\left.\left.+ 2.16663826370879999998081 9 \times 10^{-9} A\right)\right)\right) \Big/ \left(7.810063313388424519 \times 10^{18} \ln\left(1.\right.\right.$$
$$\left.+ 1.16725333333333333323000 \times 10^{-7} A\right) - 9.606688948320040104811 9 \times 10^{22} \ln\left(1.\right.$$
$$\left.+ 2.40865345252266666664534 4 \times 10^{-9} A\right) + 9.605907941988701262360 0 \times 10^{22} \ln\left(1.\right.$$
$$\left.+ 2.16663826370879999998081 9 \times 10^{-9} A\right)\Big), E0 = -\left(3.18235934152776656224348 1 \times 10^{22}\right) \Big/$$
$$\left(7.810063313388424519 \times 10^{18} \ln\left(1. + 1.16725333333333333323000 \times 10^{-7} A\right)\right.$$
$$- 9.606688948320040104811 9 \times 10^{22} \ln\left(1. + 2.40865345252266666664534 4 \times 10^{-9} A\right)$$
$$\left.+ 9.605907941988701262360 0 \times 10^{22} \ln\left(1. + 2.16663826370879999998081 9 \times 10^{-9} A\right)\right), Em$$
$$= \left(0.01000000000000000000000 0\left(1.449341527766562243481 \times 10^{21} \ln\left(1.\right.\right.\right.$$

$$+ 1.1672533333333333333323000 \times 10^{-7} A) - 3.1318990000000000000000000 \times 10^{25} \ln(1.$$
$$+ 2.4086534525226666666645344 \times 10^{-9} A) + 3.4499900000000000000000000 \times 10^{25} \ln(1.$$
$$+ 2.1666382637087999999980819 \times 10^{-9} A))) / (7.810063313388424519 \times 10^{18} \ln(1.$$
$$+ 1.1672533333333333333323000 \times 10^{-7} A) - 9.6066889483200401048119 \times 10^{22} \ln(1.$$
$$+ 2.4086534525226666666645344 \times 10^{-9} A) + 9.6059079419887012623600 \times 10^{22} \ln(1.$$
$$+ 2.1666382637087999999980819 \times 10^{-9} A))\}$$

```
> A:=6993.8*10^4;evalf(sol[4]);#Em
```

$$A := 6.99380000 \times 10^{7}$$

$$Em = 0.036003711081316326770796 41 \tag{62}$$

```
> Em:=0.036003711081316326770796 41;
```

$$Em := 0.036003711081316326770796 41 \tag{63}$$

```
> evalf(sol[3]);#E0=1.722290354;
```

$$E0 = 22.9675901768571683648064 5 \tag{64}$$

```
> E0:=22.9675901768571683648064 5;
```

$$E0 := 22.9675901768571683648064 5 \tag{65}$$

```
> evalf(sol[2]);#C
```

$$C = -5.2980799518896844142007 20 \times 10^{-20} \tag{66}$$

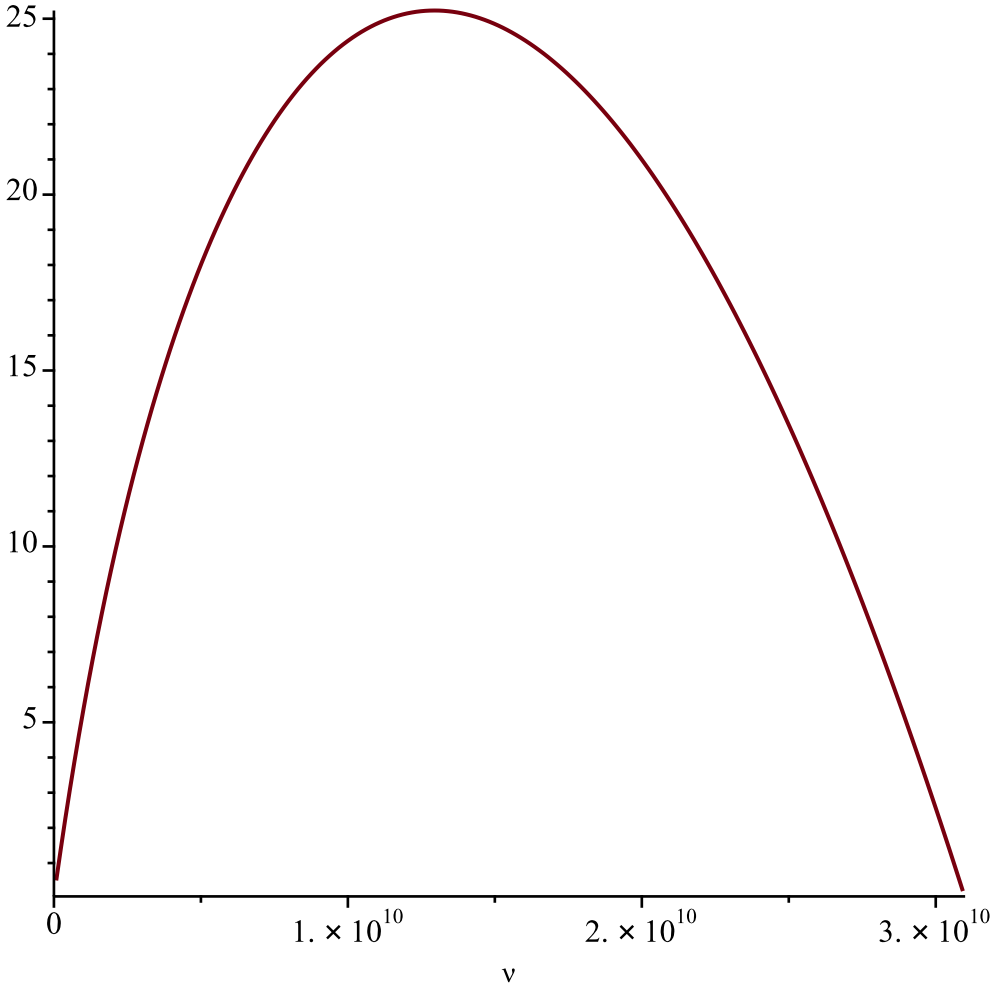
```
> C:=-5.2980799518896844142007 2*10^(-20);
```

$$C := -5.2980799518896844142007 20 \times 10^{-20} \tag{67}$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := 0.036003711081316326770796 41 + 22.9675901768571683648064 5 \ln(1 \tag{68}$$
$$+ 2.6333988266666666666433 54 \times 10^{-10} \nu) - 5.2980799518896844142007 20 \times 10^{-20} \nu^{2}$$

```
> plot(E(nu),nu=10^6..3.1*10^10);
```




```
> evalf(subs(nu=5.754173859*10^8,E(nu)));evalf(subs(nu=6.39691958*10^8,E(nu)
  ));evalf(subs(nu=7.111105195*10^8,E(nu)));
```

$$3.2590000000000000000000005$$

$$3.5900000000000000000000005$$

$$3.951629162260248123877006 \quad \textbf{(69)}$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=1*10^3,E(nu)));evalf(subs(nu=14.91*
  10^9,E(nu)));#24.9eV @ 14.91Ghz in the range [1Khz-31Ghz]
```

$$0.036003711081316326770796 41$$

$$0.036009759362969278761483 10$$

$$24.882246359040279205451 50 \quad \textbf{(70)}$$

```
> mu:=4*Pi*10^(-7);rb:= 5.292410932*10^(-11);solve((Pi*mu*E0m*rb^3)^0.5=
  2.360457908*10^(-18));
```

$$\mu := \frac{\pi}{2500000}$$

$$rb := 5.2924109320000000000000000 \times 10^{-11}$$

$$9.5207708089745780365275 66 \quad \textbf{(71)}$$

```
> h := 6.62607015*10^(-34);ec:= 1.602176634*10^(-19);he:=h/ec;rm:=5.64*10^
  (-21);Bmw:=mu*B*h/(8*Pi*rm^2);mu*B*he/(8*Pi*rm^2);E0m:=
  24.68922443288053649710286;Bmw/1.748;(8*Pi/3*mu*E0m*rb^3)^0.5/(Pi*mu*E0m*
  rb^3)^0.5;
```

$$h := 6.6260701500000000000000000 \times 10^{-34}$$

$$ec := 1.6021766340000000000000000 \times 10^{-19}$$

$$he := 4.13566769692385864616223 \times 10^{-15}$$

$$rm := 5.64000000000000000000000 \times 10^{-21}$$

$$Bmw := 3.84633523589576731552738 \times 10^{-18}$$

$$24.006936278261609671729 80$$

$$E0m := 24.689224432880536497102 86$$

$$2.200420615501011050072877 \times 10^{-18}$$

$$1.6329931618554520654648 56 \quad (72)$$

```
> (Pi*mu*E0m*rb^3)^0.5;(8*Pi/3*mu*E0m*1.478*rb^3)^0.5;(8*Pi/3*mu*E0m*rb^3)
  ^0.5/(Pi*mu*E0m*rb^3)^0.5;Bm:=Bmw/1.478;Bm/1.478;
```

$$3.801140555707731701716545 \times 10^{-18}$$

$$7.546325120098428628603176 \times 10^{-18}$$

$$1.6329931618554520654648 56$$

$$Bm := 2.602391905206879103875094 \times 10^{-18}$$

$$1.760752303928876254313325 \times 10^{-18} \quad (73)$$

```
>
```

```
> restart;#The Microwave Planck's low
```

```
> Digits:=25;
```

$$Digits := 25 \quad (74)$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, Tq\, A\, \nu + 1) + C\, \nu^2 \quad (75)$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=3.693*10^(-18);ec:=
  1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 3.693000000000000000000000 \times 10^{-18}$$

$$ec := 1.602176634000000000000000 \times 10^{-19} \quad (76)$$

```
> E(nu):=Em+E0*ln(1+A*2.8240000000000000*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.824000000000000000000000 \times 10^{-18} A\, \nu) + C\, \nu^2 \quad (77)$$

```
> A0:=16*Tq;
```

$$A0 := 2.824000000000000000000000 \times 10^{-18} \quad (78)$$

```
> E(nu):=Em+E0*ln(1+A*2.8240000000000000*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.824000000000000000000000 \times 10^{-18} A\, \nu) + C\, \nu^2 \quad (79)$$

```
> E1:=subs(nu=160.23*10^9,E(nu));
```

$$E1 := Em + E0 \ln(1 + 4.524895200000000000000000 \times 10^{-7} A) + 2.567365290000000000000000 \times 10^{22} C \quad (80)$$

```
> E2:=subs(nu=285*10^9,E(nu));
```

$$E2 := Em + E0 \ln(1 + 8.048400000000000000000000 \times 10^{-7} A) + 8122500000000000000000 C \quad (81)$$

```
> E3:=subs(nu=6.85*10^12,E(nu));
```

$$E3 := Em + E0 \ln(1 + 0.00001934440000000000000000000\, A) + 4.69225000000000000000000 \times 10^{25}\, C \quad (82)$$

```
> solve({E1=7.15,E2=12.64});
```

$$\{A = A,\ C = 1.80013636428989496098106 3 \times 10^{-23}\, E0 \ln(1. + 4.52489520000000000000000 \times 10^{-7}\, A) \quad (83)$$
$$- 1.80013636428989496098106 3 \times 10^{-23}\, E0 \ln(1. + 8.04840000000000000000000 \times 10^{-7}\, A)$$
$$+ 9.88274863995152333578603 7 \times 10^{-23},\ E0 = E0,\ Em = 0.46216076189446718205686 86\, E0 \ln(1.$$
$$+ 8.04840000000000000000000 \times 10^{-7}\, A) - 1.46216076189446718205686 9\, E0 \ln(1.$$
$$+ 4.52489520000000000000000 \times 10^{-7}\, A) + 4.61273741719937517050779 1\}$$

```
> sol:=solve({E1=7.15,E2=12.64,E3=0},{Em,E0,C,A});
```

$$sol := \{A = A,\ C = (1.00000000000000000000000 \times 10^{-16}\, (715. \ln(1. + 8.04840000000000000000000 \quad (84)$$
$$\times 10^{-7}\, A) + 549. \ln(1. + 0.00001934440000000000000000000\, A) - 1264. \ln(1.$$
$$+ 4.52489520000000000000000 \times 10^{-7}\, A))) / (5.55513471 \times 10^{8} \ln(1.$$
$$+ 0.00001934440000000000000000000\, A) - 4.68968263471 \times 10^{11} \ln(1.$$
$$+ 8.04840000000000000000000 \times 10^{-7}\, A) + 4.68412750000 \times 10^{11} \ln(1.$$
$$+ 4.52489520000000000000000 \times 10^{-7}\, A)),\ E0 = -(2.57860768777344000000000 \times 10^{12}) /$$
$$(5.55513471 \times 10^{8} \ln(1. + 0.00001934440000000000000000000\, A) - 4.68968263471 \times 10^{11} \ln(1.$$
$$+ 8.04840000000000000000000 \times 10^{-7}\, A) + 4.68412750000 \times 10^{11} \ln(1.$$
$$+ 4.52489520000000000000000 \times 10^{-7}\, A)),\ Em$$
$$= (0.64000000000000000000000 (4.003809021 \times 10^{9} \ln(1. + 0.00001934440000000000000000000\, A)$$
$$- 5.242123046875 \times 10^{12} \ln(1. + 8.04840000000000000000000 \times 10^{-7}\, A) + 9.267193750000$$
$$\times 10^{12} \ln(1. + 4.52489520000000000000000 \times 10^{-7}\, A))) / (5.55513471 \times 10^{8} \ln(1.$$
$$+ 0.00001934440000000000000000000\, A) - 4.68968263471 \times 10^{11} \ln(1.$$
$$+ 8.04840000000000000000000 \times 10^{-7}\, A) + 4.68412750000 \times 10^{11} \ln(1.$$
$$+ 4.52489520000000000000000 \times 10^{-7}\, A))\}$$

```
> A:=79904.7*10^1;evalf(sol[4]);#Em
```

$$A := 799047.0$$
$$Em = -1.97829782870395672079387 0 \quad (85)$$

```
> Em:=-1.97829782870395672079387 0;
```

(86)

$$Em := -1.97829782870395672079387\,0 \tag{86}$$

```
> evalf(sol[3]);#E0=28.87280028
```

$$E0 = 29.720763361662431074128\,13 \tag{87}$$

```
> E0:=29.72076336166243107412813;
```

$$E0 := 29.720763361662431074128\,13 \tag{88}$$

```
> evalf(sol[2]);#C
```

$$C = -1.731841026798320557990073 \times 10^{-24} \tag{89}$$

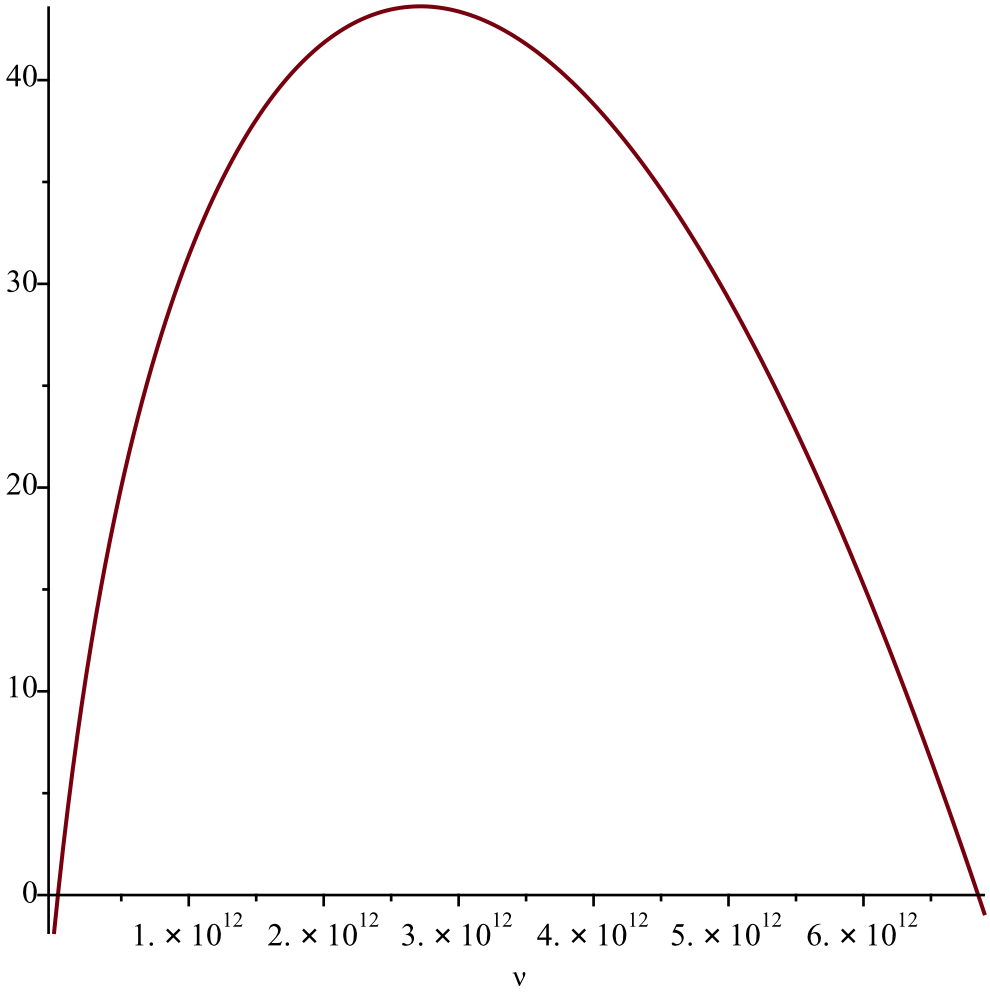
```
> C:=-1.731841026798320557990073*10^(-24);
```

$$C := -1.731841026798320557990073 \times 10^{-24} \tag{90}$$

```
> E(nu):=Em+E0*ln(1+A*2.824*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := -1.97829782870395672079387\,0 + 29.720763361662431074128\,13 \ln\left(1 + 2.25650872800000000000000 \times 10^{-12}\,\nu\right) - 1.731841026798320557990073 \times 10^{-24}\,\nu^{2} \tag{91}$$

```
> plot(E(nu),nu=10^9..6.9*10^12);
```




```
> evalf(subs(nu=285*10^9,E(nu)));evalf(subs(nu=160.23*10^9,E(nu)));evalf
  (subs(nu=5.85*10^12,E(nu)));
```

$$12.640000000000000000000\,1$$

$$7.150000000000000000000\,02$$

$$17.611354876254039343043\,45 \tag{92}$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=30.4*10^9,E(nu)));evalf(subs(nu=
  2.79*10^12,E(nu)));#43.9eV @ 15.4Ghz [30Ghz÷6.85Thz]
```

$$-1.97829782870395672079387\,0$$

$$-0.00800345982571555634891 4$$

$$43.60434402635505383696412 \quad (93)$$

```
> c := 299792458;c/(6.85*10^12)
```

$$c := 299792458$$

$$0.00004376532233576642335766422 \quad (94)$$

```
> solve(12=160.23*10^9*hmw);#The energy of CMB
```

$$7.48923422580041190788241 9 \times 10^{-11} \quad (95)$$

```
> hmw:=7.48923422580041190788241 9*10^(-11);
```

$$hmw := 7.48923422580041190788241 9 \times 10^{-11} \quad (96)$$

```
> 160.23*10^9·hmw ; 285·10^9·hmw;
   #Penzias and Wilson originally detected the Cosmic Microwave Background
   160.23÷285 GHz with an energy of ~12.64÷22.49 eV ;
```

$$12.000000000000000000000 0$$

$$21.344317543531173937464 89 \quad (97)$$

```
> # http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/h21.html
> #10.9Ghz is 3.59eV on the microwave thus we can deduce the Planck's
  constatnt for miccrowave;
> E:=hmw*nu;
```

$$E := 7.48923422580041190788241 9 \times 10^{-11} \nu \quad (98)$$

```
> hm1w=3.59/(10.9*10^9);#Posible value
```

$$hm1w = 3.29357798165137614678899 1 \times 10^{-10} \quad (99)$$

```
> 160.23*10^9 ;#Penzias and Wilson originally detected the Cosmic Microwave
  Background 160.23GHz with an energy of ~12eV ;
```

$$1.6023000000000 \times 10^{11} \quad (100)$$

```
> c := 299792458;
  160.23*10^9;
  evalf(c/(1.9*10^11));evalf(c/(10.9*10^9));
```

$$c := 299792458$$

$$1.6023000000000 \times 10^{11}$$

$$0.0015778550421052631578947 37$$

$$0.027503895229357798165137 61 \quad (101)$$

```
> hm1w=0.188977/(1.057*10^9);13.60569292/4;3.5904-3.401423;#Posible value
```

$$hm1w = 1.78786187322611163670766 3 \times 10^{-10}$$

$$3.401423230000000000000 0$$

$$0.188977 \quad (102)$$